

CO2 AT-3

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- 1) An image appears to be bright
Suitable image level transformation image negative or
logically, power law transformation

Concept :- the power law transformation modifies the intensity
using the equation

$$S = Cx^{\gamma}$$

x = input pixel intensity

S = output pixel intensity

γ = gamma value

C = constant

Expected outcome:-

- overall brightness decreases
- hidden details become visible
- Image appears more natural
- High intensity regions are preserved without saturation

A dark image lacks detail

Dark image contain most intensity values lower grey level range
making detail visible the enhancement technique is gamma correction

$$S = Cx^{\gamma}$$

Application process

1. read the image
2. choose gamma less than 1
3. Display the enhanced image
4. expected outcome

- Dark regions become brighter
- hidden objects become visible
- Bright regions remain almost unchanged

② An Image has poor contrast

When an Image has poor contrast pixel intensities occupy only a small range histogram equalization spreads the intensity values across the full gray level range

Working process

1. calculate histogram
2. compute cumulative histogram
3. Normalize the CDF

- Bright regions gain more detail
- Overall contrast improves significantly

3. An Image shows uneven brightness across regions when different regions of an image have different brightness levels. Contrast limited adaptive histogram equalization is the best enhancement technique.

Working Process

1. Divide the images into small regions
2. perform histogram equalization on each slice
3. Merge neighboring slices using interpolation

Expected outcome

- corrects non-uniform illumination
- Improves local contrast
- Preserve image details
- Reduces noise amplification
- produces a natural looking enhanced image

4. Create enhanced Image

Expected outcome

contrast increases

Image details become clear

edges become more visible

A Low contrast image requires uniformity intensity ^{distribution} ~~image~~ _{contrast}

To achieve a uniform intensity distribution Histogram equalization is the most appropriate method because it re-distributes pixel intensities across the grey level range

Working Process

Histogram

compute Image values

Normalize cumulative values

Replace each pixel with mapped intensity

effect on intensity levels

Pixel values spread uniformly

Dark regions become brighter

- Bright regions gain more detail
- Overall contrast improves significantly

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